

Marcus Loo Vergara

Oslo, Norway

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SUMMARY OF QUALIFICATIONS

7 years of experience working on real-time and performance-critical systems, including computer graphics and embedded software. M.Sc. in Computer Science from the Norwegian University of Science and Technology (2019), researching deep learning and computer vision. Self-motivated and adaptable, with a collaborative approach to technical design and engineering.

- C/C++
- Low-level optimization
- Vulkan, D3D11/12, & WebGPU
- Python
- API design
- Performance analysis
- Deep learning
- Software architecture
- Multi-platform development

EDUCATION

Norwegian University of Science and Technology, Trondheim, Norway Aug 2013 – Jun 2019
Master of Science in Computer Science.

- Specializing in computer graphics, computer vision, and deep learning.

University of California, San Diego, San Diego, USA Oct 2017 – Jun 2018
Additional Studies, Computer Science.

- Studying at UC San Diego as part of the degree.

WORK EXPERIENCE

Senior Software Developer at Hexagon, Oslo, Norway Aug 2024 – Present

Software Developer Nov 2022 – Aug 2024

Full-time R&D role focused on rendering large-scale CAD data and optimizing low-level rendering for mobile, desktop, and web.

- Extending the internal rendering abstraction with modern rendering features:
 - Compute shaders, storage buffers, uniform buffers, descriptor sets, samplers.
 - Implemented on DirectX 11, DirectX 12, Vulkan, OpenGL, WebGL, WebGPU, Metal.
- Optimizing rendering of parametric geometry, with up to 30% improvement on some devices.
- Developing functionality to convert and visualize common CAD-file formats.
- Designing a code generation solution for cross-language APIs, leveraging C as a common FFI.

Drone Software Engineer at Nordic Unmanned, Oslo, Norway Feb 2021 – Nov 2022

Full-time position developing and integrating embedded systems for bespoke maintenance and delivery drones.

- Developed drivers for an STM32 Cortex-M7 microcontroller to acquire data from onboard IMUs.
- Implemented video streaming from onboard camera, ensuring stability on a low-bandwidth link.
- Developed software modules for integrating third-party hardware over UART.
- Improved and extended the in-house simulator, including hardware-in-the-loop testing.

Junior Graphics Programmer at Funcom, Oslo, Norway Oct 2019 – Feb 2021

Full-time position as a graphics programmer working in Unreal Engine on *Dune: Awakening*.

- Researched rendering techniques and extended Unreal Engine's material system with bindless texture support for PC (DirectX 12), Xbox One (DirectX 12), and PS4 (libGNM).
- Optimized development workflows for artists and other users.
- Provided internal support for rendering issues and troubleshooting.

Graphics Software Engineering Intern at Sony Interactive Ent., San Mateo, USA Jul 2018 – Sep 2018

Internship at SIEA's Global Developer Technology Group.

- Implemented a physically based rendering and HDR demo for PlayStation 4 (libGNM and PSSL).
- Investigated tools for automatically extracting and analyzing GPU performance data.

Software Developer at Grabster, Trondheim, Norway Jul 2017 – Aug 2017
Summer job assisting a start-up with app development.

- Grabster was a food-selling marketplace app for iOS and Android.
- Implemented screen navigation and backend functionality for the Android app (*React Native*).

Teaching Assistant at the Norwegian University of Science and Technology Aug 2015 – May 2019
On-campus part-time job assisting students.

- Support and advising in solving theoretical and practical questions.
- Subjects:
 - Jan 2019 – May 2019: Computer vision and deep learning
 - Jan 2016 – May 2017: Procedural and object-oriented programming (C++)
 - Aug 2015 – Dec 2015: Computers and digital design (Assembly programming)

RELEVANT PROJECTS

Fractala Sandbox, Game project 2019 – Present
Co-owner and one of two developers of *Fractala Sandbox*, a large-scale GPU-driven physics sandbox written from scratch in C++.

- Position-based dynamics simulating more than 100,000 particles at 240 Hz.
- Supported rendering APIs: Vulkan and DirectX 11.
- Threaded rendering and detailed GPU/CPU performance metrics.
- See it in action: <https://youtu.be/AKEU9soBSvE>

Reinforcement Learning for Autonomous Vehicles, Thesis Sep 2018 – Jun 2019
Research in deep reinforcement learning, implementing methods to teach a simulated car to drive in the open-source driving simulator *CARLA*.

- Implemented a custom version of the *Proximal Policy Optimization* reinforcement learning method with *Python* and *TensorFlow*.
- Created an *OpenAI Gym*-like environment for *CARLA* to make RL research easier.
- Reduced training time through experimentation with reward functions, environment designs, and improvements to the PPO algorithm by scaling the output mean to respective limits.

MISCELLANEOUS

Seismic Map Illustration for European Standard, Commissioned Work Apr 2021
Used official seismic data from EFEHR along with geographical boundaries of municipalities in Norway to programmatically create a seismic map, printed in the official standard: *NS-EN 1998-1:2004+A1:2013+NA:2021*.

Teaching Assistant Training, Norwegian University of Science and Technology May 2019
Certificate awarded for attending a pedagogical training course for teaching assistants.

Abakus GameDev, Norwegian University of Science and Technology Aug 2014 – Jun 2019
Co-founder and leader of student organization *Abakus GameDev* – a student organization for people who enjoy making games, discussing their development, design, and the industry in general.

- Responsible for creating a socket-based framework that was used for AI competitions.
- Participated in meetings, discussing workshop ideas and advertising.
- Held a presentation on deep reinforcement learning for games.